



HUMAN EVOLUTION

Turning Back the Clock: Slowing the Pace of Prehistory

New work suggests that mutations arise more slowly in humans than previously thought, raising questions about the timetable of evolutionary events

Timing is everything, especially in human evolution. For the past 45 years, researchers have used the number of mutations in DNA like a molecular clock to date key chapters in the human evolutionary story, such as the dawn of humankind millions of years ago and the exodus of modern humans from Africa in the past 100,000 years.

Now it seems that the molecular clock ticks more slowly than anyone had thought, and many dates may need to be adjusted. Over the past 3 years, researchers have used new methods to sequence whole human genomes, allowing them to measure directly, for the first time, the average rate at which new mutations arise in a newborn baby. Most of these studies conclude that the mutation rate in humans today is roughly half the rate that has been used in many evolutionary studies since 2000. "Together, these papers make a convincing case that the human sequence mutation rate is substantially less than the one previously used," says Harvard University population geneticist David Reich, co-author of one recent study. "As a result, genetic estimates of dates for ancient events are going to be older than previously reported."

The question now is how much older? Three new studies have taken a stab at providing an answer, trying to apply the slower mutation rates to major events in human evolution. In the past month, they have published a range of dates that sometimes fit with evidence from the fossil record—and that are sometimes way off, particularly for events further back in time. Some researchers suggest that the molecular clock may not

keep steady time across primates, and that it also has slowed over millions of years. This is confusing researchers who rely on those rates to study genetic diseases and the evolution of humans and primates.

"The mutation rates are so up in air," said paleogeneticist Svante Pääbo of the Max Planck Institute for Evolutionary Anthropology in Leipzig, Germany, in August, when his team published big margins of error—from 170,000 to 700,000 years ago—for the date when our ancestors split from Neandertals and their close cousins, the Denisovans. As a result, the timing of some events in human origins is now "very murky," says paleoanthropologist Chris Stringer of the Natural History Museum in London. The ambiguity in the mutation rate affects a host of evolutionary and disease-related analyses, says paleoanthropologist John Hawks of the University of Wisconsin, Madison: "We can't figure out how things happened if we don't know when they happened."

Fossil time

For the past 15 years, researchers have estimated the speed of the molecular clock by counting the mutational differences between humans and primates in matching segments of DNA, then using different species' first appearances in the fossil record to estimate how long it took those mutations to accumulate. For example, the fossils of the oldest known orangutan ancestor are about 13 million years old, so DNA differences between humans and orangutans had about that long to accumulate. By doing similar calculations in many segments of DNA in various

Time warp. The molecular clock used to date events in our prehistory may run more slowly than had been thought, and at variable speeds.

primates, researchers calculated an average rate of about one mutation per billion base pairs per year for humans and other apes.

When researchers plugged this rate into their equations, most got dates between 4 million and 6 million years ago for the split between the ancestors of humans and chimps. That dovetails pretty well with fossils identified as the earliest known hominins, or members of the human family, such as *Sahelanthropus*, which lived 6 million to 7 million years ago; *Orrorin*, which lived 6 million years ago; and *Ardipithecus*, which lived 4.4 million to 5.8 million years ago (*Science*, 15 February 2002, p. 1214).

As the rate became widely accepted, researchers used it to date milestones such as when the first modern humans migrated out of Africa—less than 70,000 years ago—and when they parted ways with Neandertals and Denisovans.

But this method of calculating the mutation rate has drawbacks. For starters, it assumes that the fossil dates accurately record the first appearance of a species, but that can change with a new find. Second, there are no fossils of our closest living relatives: chimps and gorillas. Third, the method assumes that species split at the same time as their genes diverged, but in fact, genetic separation can be millions of years earlier than species divergence. Finally, the method assumes that mutation rates are similar across apes, although factors such as generation time—the average number of years between generations—affect the rate.

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Podcast interview
with Ann Gibbons
(http://scim.ag/pod_6104).

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	Fossils	Old Mutation Rate	New Mutation Rate
Human-orangutan split	9 million–13 million years ago (<i>Sivapithecus</i>)	13 million–14 million years ago	34 million–46 million years ago
Human-chimp split	4.1 million–7 million years ago (<i>Sahelanthropus</i> , <i>Orrorin</i> , <i>Ardipithecus</i> , and <i>Australopithecus</i>)	4 million–7 million years ago	8 million–10 million years ago
Human-Neandertal split	350,000–600,000 years ago (<i>Homo heidelbergensis</i>) 200,000 years ago (Neandertals)	250,000–350,000 years ago	400,000–600,000 years ago
Out-of-Africa migration	125,000–80,000 years ago (archaic <i>Homo sapiens</i>)	Less than 70,000 years ago	90,000–130,000 years ago
Seeking synchronicity. Some dates derived using the new, slower mutation rate match fossils—and some don't.			
		Match between fossils and molecular date	
		Mismatch between fossils and molecular date	

The first sign that something was amiss came in 2003, when a study tracking genes that cause hemophilia, muscular dystrophy, and other diseases in parents and children found slower mutation rates than expected.

With the recent advent of high-throughput sequencing methods, geneticists finally have been able to sequence enough whole genomes to calculate directly the number of mutations between trios of two parents and their child in large numbers of families. Eight studies in the past 3 years (and the 2003 study) have estimated a slower mutation rate, according to a review published online on 11 September in *Nature Reviews Genetics* by geneticists Aylwyn Scally and Richard Durbin of the Wellcome Trust Sanger Institute in Hinxton, U.K. Several studies sequenced the complete genome, such as one in August that measured the mutation rate in 78 trios of Icelandic parents and children. Other papers report the mutation rate in disease-causing genes, including one study that sequenced genes in 14,000 individuals.

The Icelandic study found that on average, every newborn baby has 36 spontaneous new mutations, those not inherited from either parent. These rare, so-called *de novo* mutations are just a tiny proportion of the 3 billion nucleotide bases in the genome, but this is the source of all variation in humans. It is the raw material for all human evolution—for better or worse, because most new mutations are deleterious. “All of the diversity in the human genome was once upon a time *de novo* mutation,” says geneticist Kári Stefánsson of deCODE Genetics in Reykjavík, co-author of two new studies in Icelanders. “We have evolved through these new mutations.”

Remarkably, all the studies got about

the same rate: 1.2×10^{-8} mutations per generation at any given nucleotide site. That’s about 1 in 2.4 billion mutations per site per year (assuming an average generation time of 29 years)—and that’s less than half of the old, fossil-calibrated rate. “It really does look good; all the studies are pointing in the same direction, averaged over many individuals,” says evolutionary biologist Michael Lynch of Indiana University, Bloomington, co-author of one of the studies on the mutation rate.

If geneticists know the average rate, they can measure whether it is higher in genes that play a part in diseases such as autism, schizophrenia, or prostate cancer that are on the rise, Stefánsson says. And then they can begin to tease out what factors might change the rate in those genes.

Pushing back time

A slower-ticking molecular clock also has major implications for evolution. For example, the slow clock suggests that the ancestors of modern humans and Neandertals diverged about 400,000 to 600,000 years ago, rather than 272,000 to 435,000 years ago. This fits nicely with fossils of *Homo heidelbergensis*, which date between 350,000 and 600,000 years ago and are thought to be ancestral to Neandertals. Scally and Durbin also revised the dates for modern human evolution, such as pushing back the timing of a dramatic population bottleneck from 100,000 to 120,000 (rather than 60,000 to 80,000) years ago, and the emergence of modern humans out of Africa to 90,000 to 130,000 (rather than less than 70,000) years ago.

Researchers who study fossils and stone tools older than 100,000 years in the Middle East and North Africa say this boosts the

the Neandertal dates seem on the early side to Stringer: “We probably do not have clear Neandertal features before about 400,000 years ago—or at a real push, 500,000 years,” he says. “So, I do not see good reason—yet—for major revision” of the dates for Neandertal and modern human origins.

What’s more, the new rate slows the pace of evolution in apes to a downright crawl (see table). It puts the split of humans and chimpanzees, for example, at between 8.3 million years ago and 10.1 million years ago—far too early, given current fossil dates. The split of the lineages leading to orangutans and the African apes, including humans, goes back to 34 million to 46 million years ago, Reich says. “A human-orangutan split at 40 million years is absolutely crazy,” says paleoanthropologist David Begun of the University of Toronto, St. George, in Canada, who notes that fossils of likely orangutan ancestors date from 9 million to 13.9 million years ago.

Recognizing these problems, Scally and Durbin propose a fix: They endorse a 30-year-old idea that the mutation rate was faster early in primate evolution, then slowed in the African apes, and perhaps slowed even more in human evolution—the so-called hominoid slowdown. This idea is backed by evidence that as body size gets bigger in various mammals, their generation times get longer, slowing per-year mutation rates, Scally says. It’s an intriguing idea that leads to a better fit between fossils and Scally’s dates. “Some kind of slowdown certainly occurred in apes,” Reich says. But he cautions that “the magnitude of slowdown required to reconcile these dates is extreme.”

Indeed, a separate study recently challenged the link between bigger bodies and longer generation times. After a decade of

case that these ancient people might be ancestral to modern humans rather than a failed exodus out of Africa. But others who have tied the bottleneck or out-of-Africa migration to events such as a major volcanic eruption about 70,000 to 75,000 years ago are likely to be less pleased. “Some people will be happy with these rates, and some will not,” Hawks says.

Push back a little further in time, and the fit with fossils becomes strained. For example,

work recording the age of parents at the birth of 226 chimpanzees and 105 gorillas in the wild in Africa, Linda Vigilant of the Max Planck Institute for Evolutionary Anthropology in Leipzig, Germany, and an international team reported in August that gorillas have a shorter generation time—about 19 years—than their smaller human and chimp cousins, who average about 29 years (thanks to older fathers) and 25 years, respectively.

Vigilant's team, whose work was published in the *Proceedings of the National Academy of Sciences*, went a step further, multiplying their new generation times by the new, slower mutation rate to date events in evolution. Their result also pushes back the human-chimp split to 7 million to 13 million years ago, which just fits with the oldest fossils, Vigilant says. But those dates assume that generation times haven't changed in millions of years. "A big question is, what were generation times in the past?" Vigilant says.

Reich, meanwhile, is a co-author on yet another study that used a different method and got an intermediate rate. He says it's possible that the new methods aren't picking up all the mutations, and so could be getting an artificially slow rate. So he, Stefánsson, and graduate student James Sun of the Massachusetts Institute of Technology in Cambridge analyzed mutation rates in 2500 microsatellites—small pieces of DNA that vary in the number of times they repeat—in 85,000 people in Iceland. Microsatellites have a higher rate of mutation than DNA nucleotides, so it is easier to detect all their new mutations. The team came up with a way to convert the microsatellite mutation rates back to a base pair mutation rate, which they estimate at 1 in 1.2 billion to 1 in 2.0 billion per year.

That's also a better fit with fossils, putting the split between humans and chimpanzees at about 3.7 million to 6.6 million years ago. This would create a different kind of problem, however: The oldest fossils vying to be the earliest members of the human family, such as *Sahelanthropus* at 6 million to 7 million years ago, might be too old to be hominins. So no matter how researchers calculate the mutation rate directly, they can't accommodate all the fossil dates.

Rate adjustment

All these complicating factors show that researchers are still exploring what alters the mutation rate, and by how much. Yet another twist may be the age of fathers at conception. In a landmark study of 78 trios of parents and offspring in Iceland, Stefánsson and Augustine Kong, also of deCODE Genetics,

found that older fathers pass on more mutations to their offspring, because mistakes are made as aging DNA is copied to produce sperm over men's lifetimes. (Older mothers don't pass on more mutations, because eggs are all produced before birth.) The effect is large, causing two extra mutations per year of father's age. That means that 36-year-old dads pass on twice as many mutations as 20-year-old dads, according to a report in *Nature* in August. So an increase in the average age of fathers might have sped up the molecular clock. "It's difficult to extrapolate back like this," Stefánsson says. "You have to recognize that there are going to be flawed estimates."

Other researchers are trying to explore what affects the mutation rate by studying parts of the genome with rates that are much above or below the average. "There is a lot of variability in the rate at which DNA evolves from one part of the genome to another," says geneticist Katherine Pollard of the University of California, San Francisco, although she adds that such hot spots are such a small percentage of the chimp and human genomes that they should not affect the average mutation rate.

And in a forthcoming paper, Lynch and his colleagues show that big genomes and big populations both slow the mutation rate in model organisms such as bacteria, yeast,

plants, and fruit flies. This underscores how risky it is to assume that the mutation rate was constant in humans and apes, because so little is known about what might speed up or slow the clock over great sweeps of time. A bottleneck that cut the number of breeding adults to 10,000 before modern humans swept out of Africa would be expected to speed up the rate, Lynch says. But did the rapid expansion of modern human populations after the invention of agriculture then slow the rate? To be surer about extrapolating the rate across the apes, researchers need to calculate the mutation rate directly in chimp and gorilla parents and their offspring, as they have done for humans, Lynch says. But even when that work is done, the modern rate in these small, endangered ape populations may be faster than it was in larger ancestral populations.

Thus, although both geneticists and paleoanthropologists are attuned to the fact that the human mutation rate today is slower than previously believed, it's still unclear how much slower it is, and what that means for timing events in our past. For now, "the question remains open—what is the true value of the mutation rate," Reich says. And "to do evolutionary analysis, you want to calibrate our clock properly."

—ANN GIBBONS



Just in time? Different mutation rates make this 6-million to 7-million-year-old fossil of *Sahelanthropus* too young, too old, or just the right age to be an ancestor of humans.